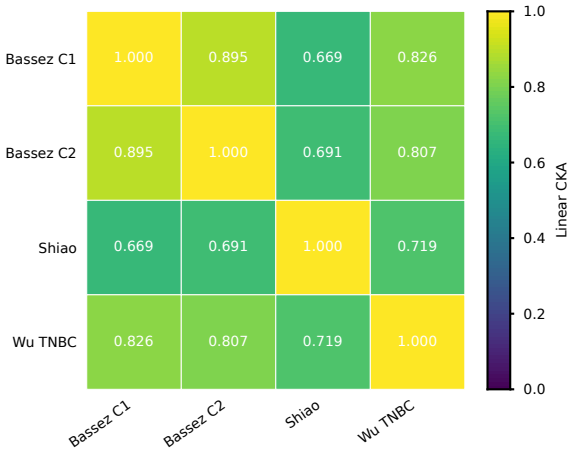
Supplementary Figure 1 | Cross-dataset W concordance and pooled W-space structure

A Cross-dataset W concordance



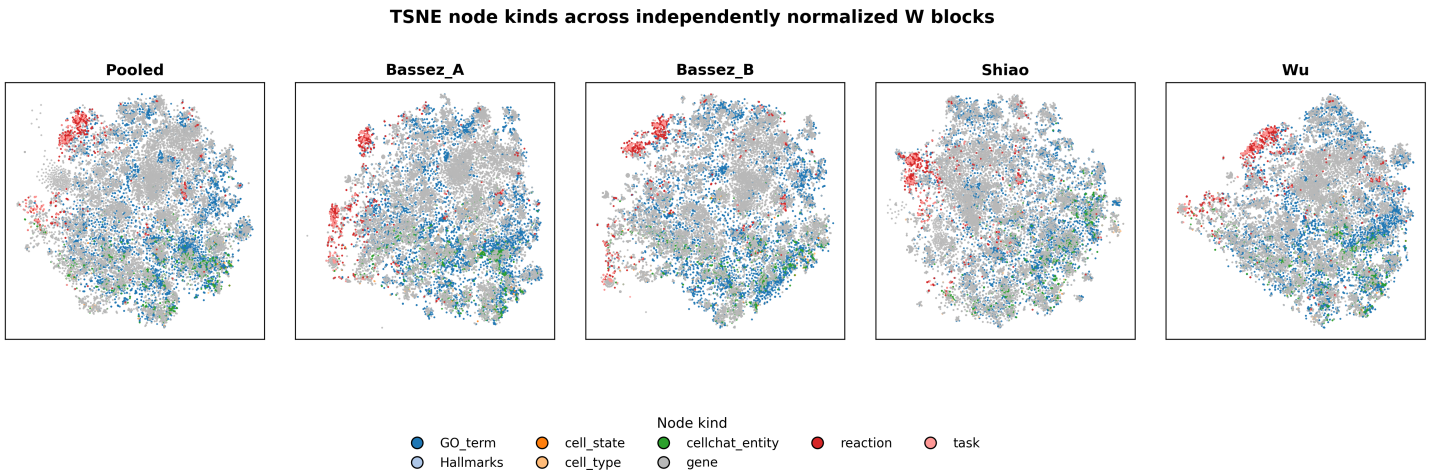
Within-dataset factor diagnostics

Dataset	Factors	Eff. rank	Med. |r|	P95 |r|	Condition
Bassez C1	40	22.2 (56%)	0.167	0.467	10.23
Bassez C2	40	22.9 (57%)	0.148	0.466	8.90
Shiao	80	56.8 (71%)	0.034	0.283	7.35
Wu TNBC	30	15.8 (53%)	0.239	0.551	7.87

Concordant W structure with limited factor redundancy

CKA quantifies shared nonlinear structure; effective rank and factor correlations quantify within-run redundancy.

B t-SNE node kinds across independently normalized W blocks



C t-SNE pooled clusters across independently normalized W blocks

